

Saransh Agrawal

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🌐 [Website](#)

🐙 [GitHub](#)

📍 Palo Alto, CA

Education

Texas A&M University *Master of Science in Data Science: (GPA: 3.7/4.0)* College Station, TX, May 2025

Courses: Parallel Computing, Database Mgmt., Natural Language Processing, Reinforcement Learning, Data Stream & Algo.

Manipal Institute of Technology *Bachelors in Electronics and Instrumentation Engineering* Aug 2021

Courses: Data Structures & Algorithms, Microprocessors, Digital Image Processing, Embedded Systems, Digital System Design

Technical Skills

Languages: Python, Go, Rust, C++, C, SQL, NoSQL(Mongo), Bash/Shell Scripting

Tools: Docker, Jenkins, Terraform, Kubernetes, PyTorch, Numpy, Scikit-Learn, MCP, Apache Spark, Hadoop, Git

Cloud: AWS, EC2, Lambda, ECS/EKS, S3, Redshift, EMR, RDS, DynamoDB, Sagemaker, Bedrock

Certificates: [AWS Solutions Architect - Professional](#)

Work Experience

Research Assistant

Aug 2024 – May 2025

FLAIR Lab, Texas A&M University

College Station, TX

- Engineered a distributed, multi-node/multi-GPU training pipeline for LLMs (Llama3, Mistral-7B) on a SLURM HPC cluster using PyTorch, DeepSpeed, and Accelerate, scaling training throughput by 4x.
- Reduced inference latency by 21% for a long-context multimodal model (LLaVA) by implementing Streaming LLM and Cache Merging techniques, increasing effective context length by 20%.
- Architected and built a scalable framework to automate LLM benchmarking (MMLU) across diverse GPU configurations, managing job scheduling and resource allocation with SLURM.

Software Development Engineer

July 2021 – July 2023

Viewzen Labs

- Architected a scalable MLaaS platform that streamlined deployment for 50+ users, proven to reliably handle over 80 concurrent training jobs during peak demand.
- Designed a robust microservices architecture using REST APIs, Kafka, and Docker to ingest and process over 5M data points daily with 99.9% uptime.
- Optimized a high-volume data transformation pipeline by 20% by re-engineering critical components in C++ and implementing concurrent processing.
- Developed and deployed an end-to-end prediction system to identify at-risk users from a highly imbalanced dataset, achieving an F1-score of 0.85 in production.
- Drove the adoption of engineering best practices by mentoring 2 junior developers on system design and automated CI/CD pipelines with Jenkins.

Machine Learning Engineer Intern

Feb 2021 – June 2021

Centre for Digital Financial Inclusion

- Engineered and deployed an end-to-end Speech-to-SQL microservice using Docker, Flask, and PostgreSQL to power a natural language interface for a data dashboard.
- Enabled real-time, voice-activated data visualization, achieving a 90% success rate in generating valid SQL queries from speech and delivering charts in under 4 seconds.
- Achieved a 17% Word Error Rate (WER) on a custom Indian Accent dataset by fine-tuning a DeepSpeech ASR model, ensuring high-fidelity transcription for the query generator.

Publications

- [1] Saransh Agrawal and Kuan-Hao Huang. Selective Amnesia – Constrained Unlearning for Large Language Models via Knowledge Isolation. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025) at ACL*.
- [2] Ren-Wei Liang, Chin-Ting Hsu, Chan-Hung Yu, Saransh Agrawal, Shih-Cheng Huang, Shang-Tse Chen, Kuan-Hao Huang, and Shao-Hua Sun. Adaptive Helpfulness-Harmlessness Alignment with Preference Vectors. In *European Chapter of the Association for Computational Linguistics (EACL) 2026*

Projects

- **Satellite-based Crop Monitoring System (SCMS)** — Python, FastAPI, Docker, React.js, MongoDB
 - Led backend system design for a real-time monitoring platform, architecting microservices with FastAPI and Docker to ingest and process terabytes of satellite imagery.
 - Deployed a forecasting microservice using an LSTM model, achieving 81% prediction accuracy and enabling data-driven agricultural insights via a React dashboard.
- **Graph-Based Ranking & Semantic Search Engine** — Go, Python, Apache Arrow, Sentence-Transformers
 - Engineered an end-to-end semantic search engine in Go and Python to index and serve over 70k research papers from the ACL Anthology. Boosted search relevancy by 30% over a keyword baseline by implementing a hybrid ranking system combining PageRank and Sentence-Transformers.